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NMAM INSTITUTE OF TECHNOLOGY, NITTE

(An Autonomous Institution affiliated to VTU, Belagavi)

Fifth Semester B.E. (E&E) (Credit System) Degree Examinations

Supplementary Examinations - December 2025

21EE504/20EE504/18EE504 - POWER ELECTRONICS

Duration: 3 Hours

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II** each and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain thyristor firing circuit using UJT with neat diagram.	8	L*2	1	1, 2
	b) Derive an expression for rms value of output voltage in case of phase-controlled AC voltage controller.	8	L2	2	1, 2
	c) A single-phase AC voltage controller feeds a resistive load of 20 Ω from a 230 V, 50 Hz supply. For a firing angle of 60°, calculate the rms value of output voltage and load power.	4	L3	2	1, 2
2.	a) With neat diagram, explain forced commutation technique of thyristor	6	L2	1	1, 2
	b) Explain turn-on and turn-off characteristics of thyristor.	7	L2	1	1, 2
	c) Explain on-off control of AC voltage controller and compare it with phase control.	7	L2	2	1, 2
3.	a) Explain V-I characteristics of thyristor, with a neat diagram.	6	L2	1	1, 2
	b)	7	L2	1	1, 2
	c) With neat diagram, explain single-phase full wave AC voltage controller.	7	L2	2	1, 2
Unit – II					
4.	a) With neat diagram explain the principle of step-down chopper with R load.	7	L2	3	1, 2, 3
	b) With neat diagram, explain the operation of a three-phase full wave converter with R load.	6	L3	4	1, 2, 3
	c) A single-phase full converter feeds power to RLE load with $R = 6 \Omega$, $L = 6 \text{ mH}$ and $E = 60 \text{ V}$. The ac source voltage is 230 V, 50 Hz. For continuous conduction, find the average value of load current for a firing angle delay of 50°.	7	L2	4	1, 2, 3
5.	a) Discuss the operation of four Quadrant chopper.	7	L2	3	1, 2
	b) For a type-A chopper feeding an RLE load, obtain the expression for maximum average current through the thyristor when the load current remains constant.	7	L2	3	1, 2, 3
	c) A single-phase full converter bridge is connected to RLE load. The source voltage is 230 V, 50 Hz. The average load current of 10 A is continuous over the working range. For $R = 0.4 \Omega$ and $L = 2 \text{ mH}$, compute the firing angle delay for $E = 120 \text{ V}$.	6	L3	4	1, 2, 3
6.	a) A step-up chopper has an input voltage of 150 V. The required output voltage is 450 V. The thyristor has a conducting time of 150 μs . Calculate the chopping frequency of operation.	6	L2	3	1, 2, 3
	b) Discuss with neat diagram and waveforms the working of step-up chopper with R Load.	7	L3	3	1, 2, 3
	c) A single-phase half-wave controlled rectifier supplied from 230 V AC, operates at $\alpha = 60^\circ$. If the load resistor is 10 Ω , determine: i) Power absorbed by the load (P_{dc}) ii) Power drawn from the supply (P_{ac}) iii) Power factor at the AC source	7	L2	4	1, 2, 3

P.T.O.

Unit – III

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| 7. | a) | Explain the parameters based on which performance of inverter is evaluated. | 7 | L2 | 5 | 1, 2, 3 |
| | b) | With neat waveform and equations, explain Single Pulse Width Modulation (PWM) | 6 | L1 | 5 | 1, 2, 3 |
| | c) | With neat diagram, explain single-phase full bridge inverter. | 7 | L2 | 5 | 1, 2, 3 |
| 8. | a) | Explain Multiple Pulse Width Modulation technique with waveforms and compare it with Single PWM. | 7 | L2 | 5 | 1, 2, 3 |
| | b) | Explain the working of single phase voltage source inverter. | 7 | L2 | 5 | 1, 2, 3 |
| | c) | A single-phase full bridge inverter is fed from a 220V DC source. If the inverter operates in 180° mode, calculate the RMS value of the fundamental output voltage and discuss the effect of harmonics. | 6 | L2 | 5 | 1, 2, 3 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
